FINAL PROJECT REPORT

Global AI Adoption in Education

Data Analytics with Tableau

Individual Project

Date: 30 September 2026

A descriptive analysis of global AI adoption in education using a structured dataset, Tableau visualizations, dashboards and a data story.

1. Introduction

1.1. Project overviews

The project, “Global AI Adoption in Education”, analyzes how artificial intelligence is being used in education across countries, regions and communities. The analysis uses the provided Global AI in Education.csv dataset and Tableau to transform raw records into interactive visualizations, dashboards and a story.

The project focuses on AI usage by students and teachers, school AI adoption, rural and urban usage, daily AI usage, AI tool distribution, internet penetration and related education indicators. The dataset contains 1,360 records covering 10 countries, 6 regions and the period 2015–2026.

1.2. Objectives

- Identify patterns and trends in AI usage in education over time.
- Compare AI adoption and usage across countries and global regions.
- Compare rural and urban AI usage and identify the gap between them.
- Understand the distribution of commonly recorded AI tools.
- Examine school AI adoption levels and their distribution.
- Use Tableau dashboards and a story to communicate findings clearly.

2. Project Initialization and Planning Phase

2.1. Define Problem Statement

Students and educators are trying to use AI tools for learning and teaching, but AI access and usage vary across countries and between rural and urban communities. Internet connectivity and digital infrastructure are not uniform, which makes it difficult to understand where AI adoption is growing and where gaps remain.

2.2. Project Proposal (Proposed Solution)

The proposed solution is to analyze global AI adoption in education using the provided structured dataset and Tableau. The project converts raw education and AI adoption data into clear worksheets, dashboards and a narrative story.

Objective: Identify trends and differences in AI usage, school adoption, rural/urban usage, AI tool distribution and internet connectivity.

Scope: 10 countries, 6 regions and the period 2015–2026 represented in the dataset. The analysis is descriptive and visualization-focused.

Impact: The analysis makes differences in AI usage and adoption easier to communicate and supports data-informed discussion of digital access and education technology.

2.3. Initial Project Planning

Sprint	Functional Requirement	User Story / Task	Points
Sprint-1	Project Initialization	Define problem statement and project objectives	2
Sprint-1	Data Collection & Preprocessing	Collect dataset and inspect data quality	3
Sprint-2	Data Cleaning & Modeling	Validate fields, missing values and calculations	3
Sprint-2	Data Visualization	Develop Tableau worksheets for business questions	4
Sprint-3	Dashboard & Story	Combine worksheets into dashboards and story	4

3. Data Collection and Preprocessing Phase

3.1. Data Collection Plan and Raw Data Sources Identified

The primary raw data source is the project-provided “Global AI in Education.csv” file. It contains 1,360 rows and 16 variables and covers 10 countries, 6 regions and 2015–2026. The source was provided as a CSV file for the project.

The analysis uses one primary structured data source. The original CSV was preserved as the source file and the prepared workbook/data was used for Tableau analysis.

3.2. Data Quality Report

Issue	Severity	Resolution
441 missing values in government_ai_policy	Moderate	Retain as missing/unknown; avoid assigning unsupported policy values.
No duplicate records detected	Low	No removal required; retain all 1,360 records.
Numeric/categorical Tableau field types need validation	Low	Validate field roles and data types before calculations and charts.

3.3. Data Exploration and Preprocessing

- Checked the dataset structure, missing values and duplicate records.
- Validated year, month and usage/adoption fields as numeric measures.
- Validated country, region, top_ai_tool, government_ai_policy and ai_in_curriculum as categorical fields.
- Used country, region and year groupings to calculate averages and percentages for Tableau.
- Created an AI adoption level classification for school adoption.
- No column splitting or merging was required because the source columns were already structured for analysis.

4. Data Visualization

4.1. Framing Business Questions

- What is the trend of rural and urban AI usage over time?
- How does school AI adoption compare across global regions?
- Which countries have the highest average daily AI usage?
- How is AI adoption distributed across low, moderate and high levels?
- Which AI tools appear most frequently in the education dataset?
- How does internet penetration relate to school AI adoption?
- How do rural and urban AI usage differ within countries?
- How does AI usage differ across countries on a daily basis?

4.2. Developing Visualizations

The Tableau workbook contains multiple visualizations designed to answer the business questions. The following figures are the supplied Tableau visualizations used in the project documentation.

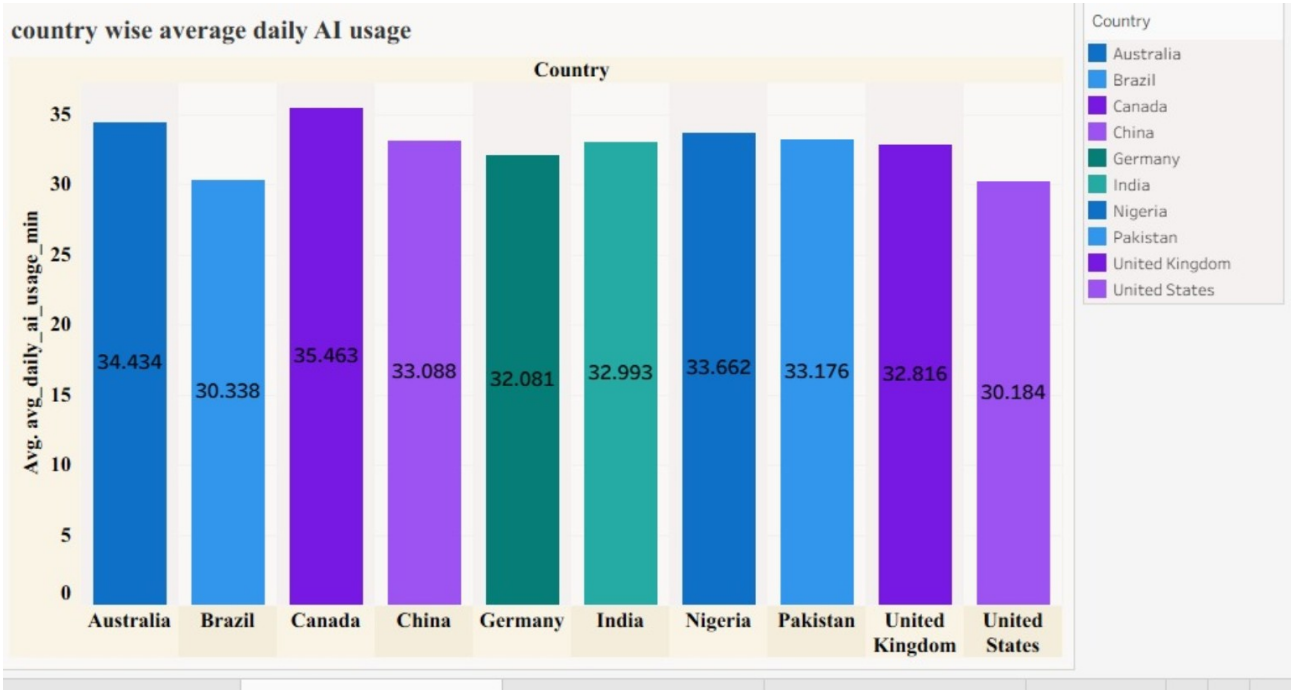


Figure 1. Country-wise average daily AI usage.

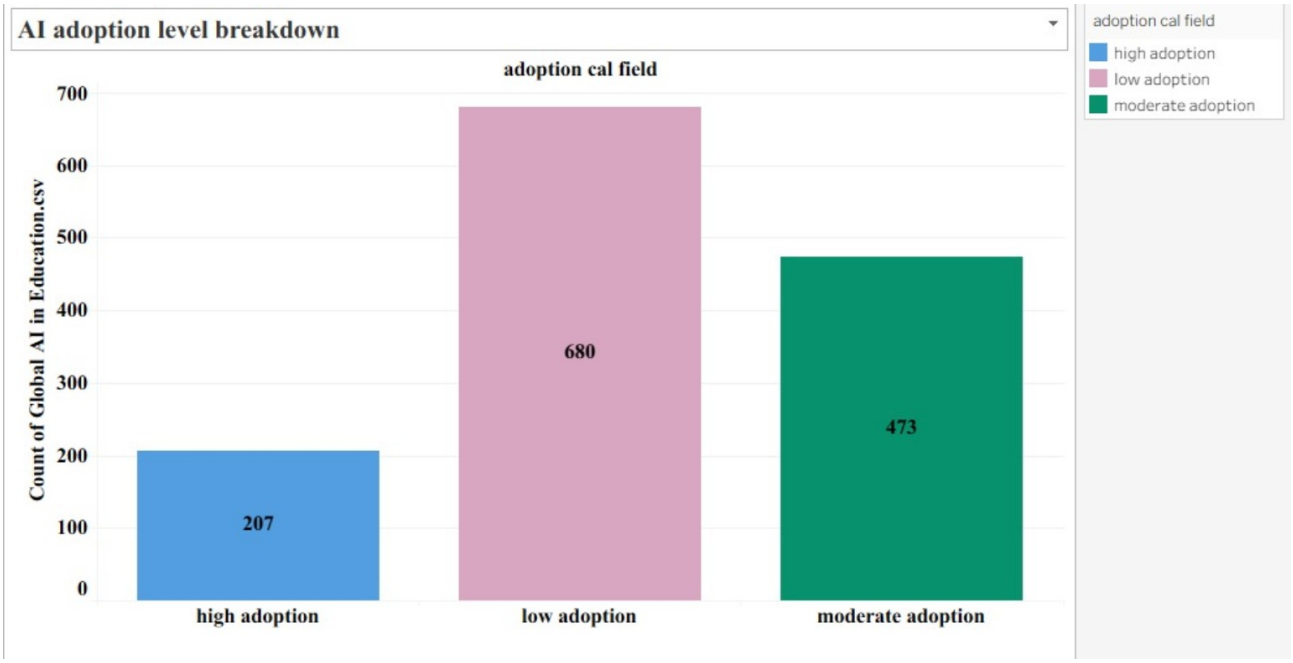


Figure 2. AI adoption level breakdown.

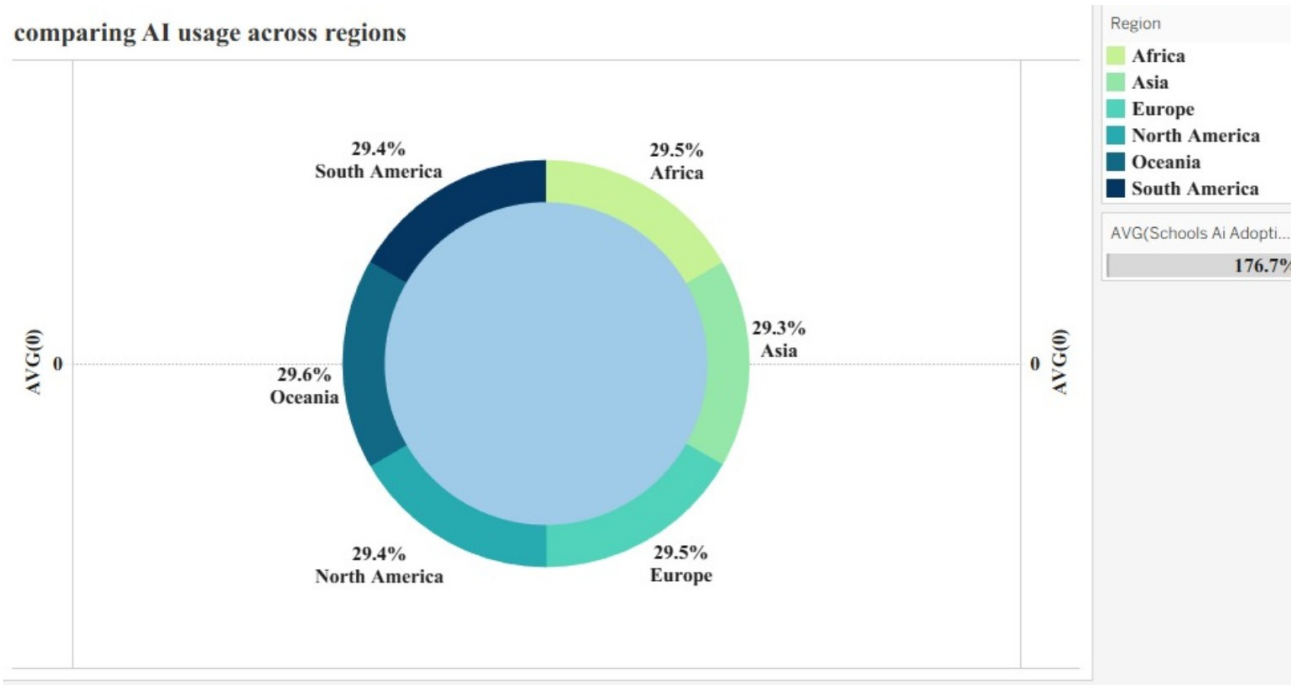


Figure 3. Comparing AI usage across regions.

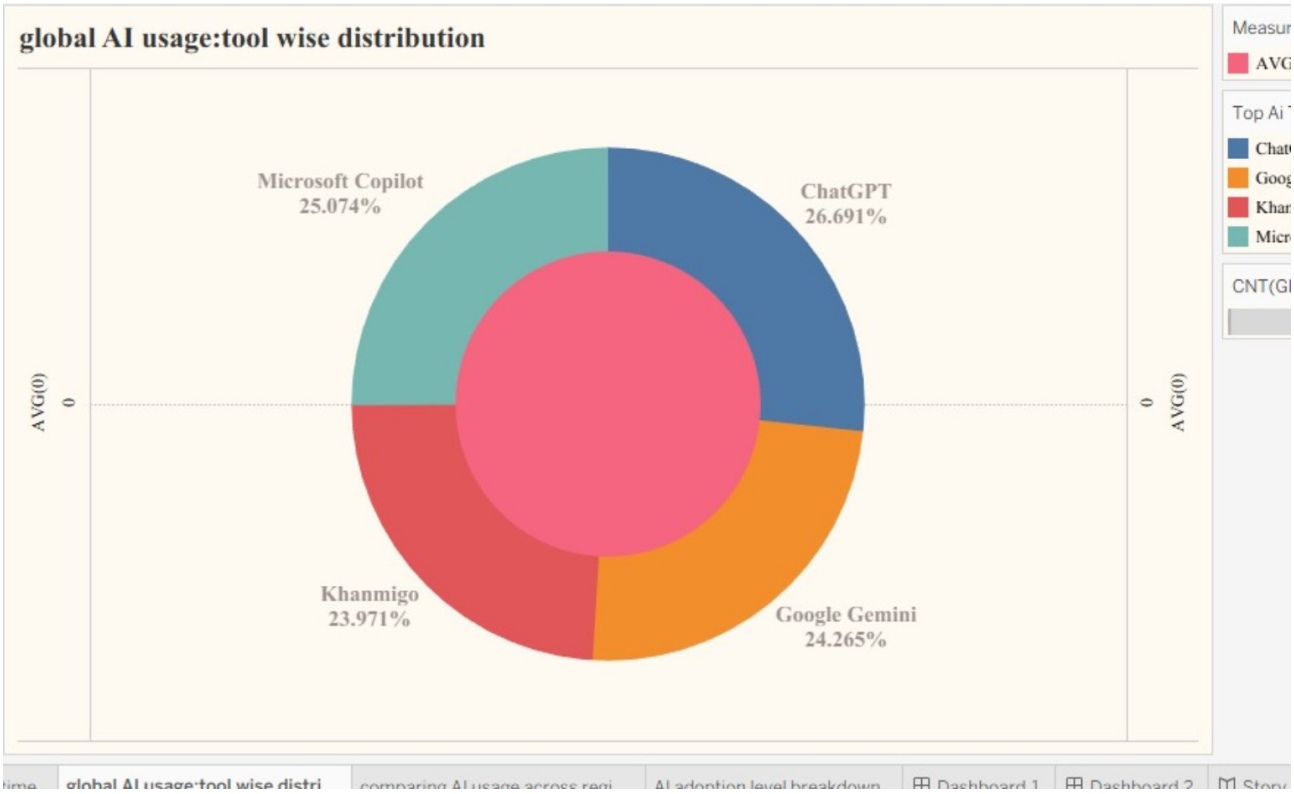


Figure 4. Global AI usage: tool-wise distribution.

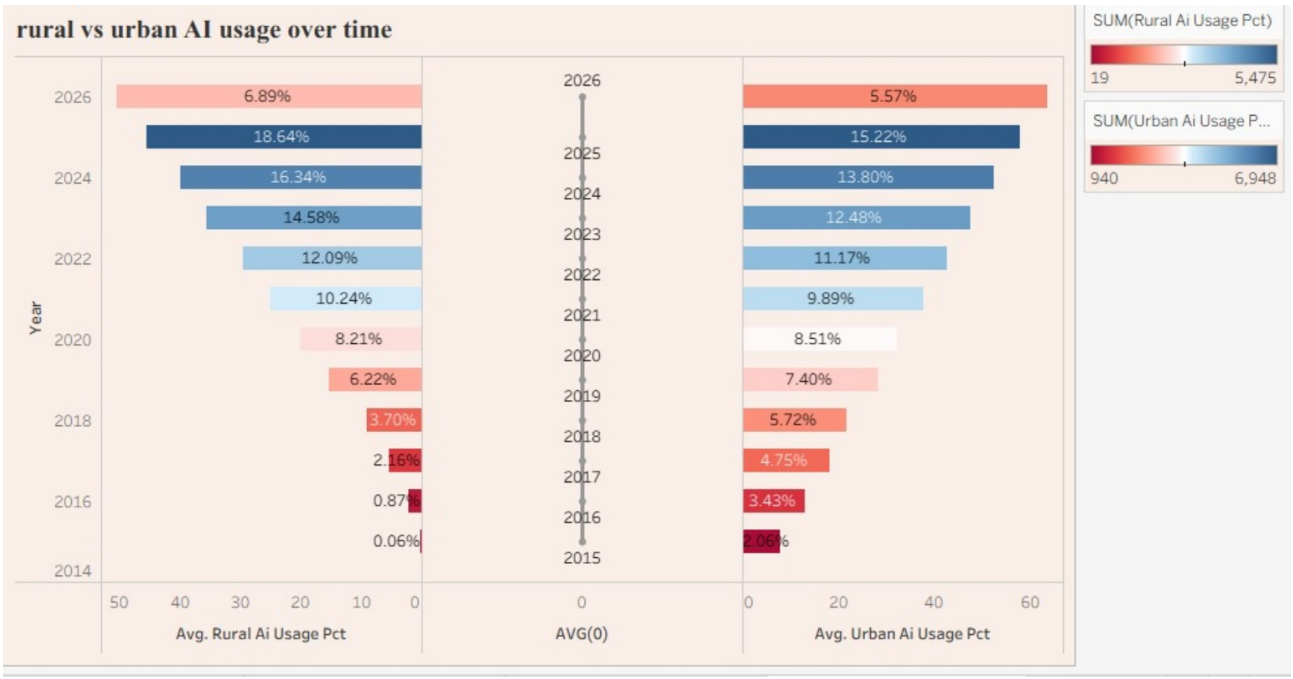


Figure 5. Rural vs urban AI usage over time.

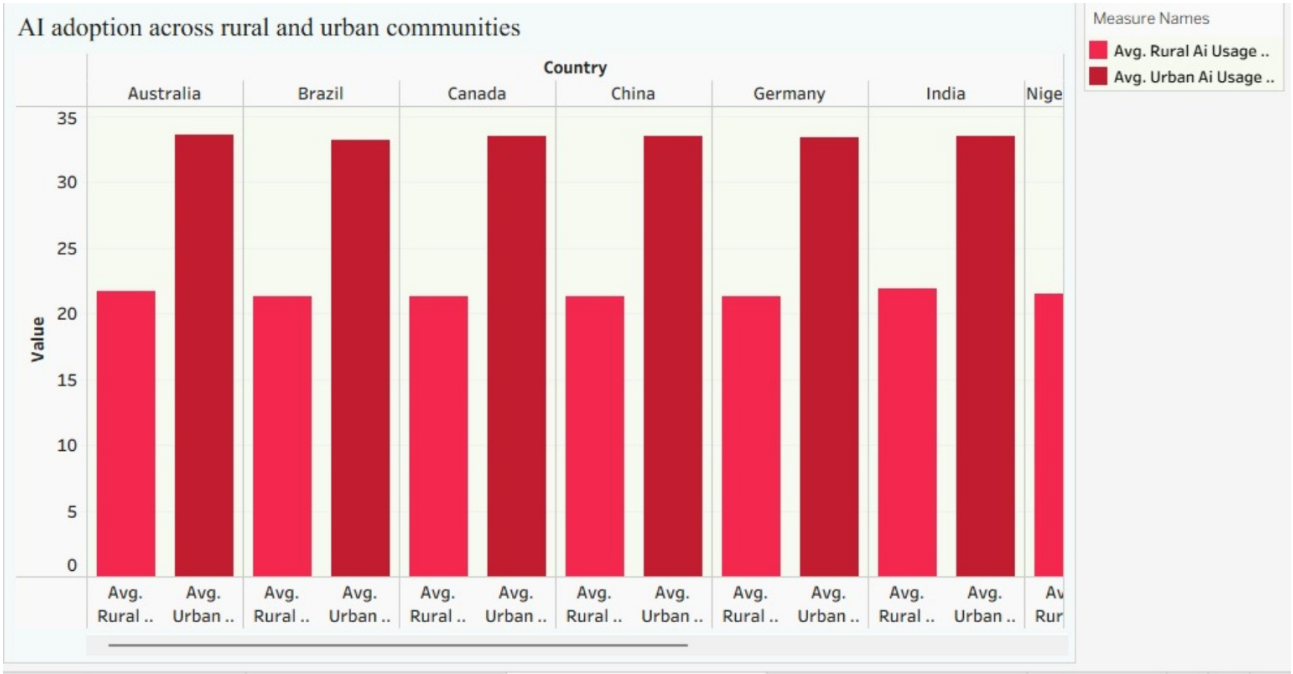


Figure 6. AI adoption across rural and urban communities by country.

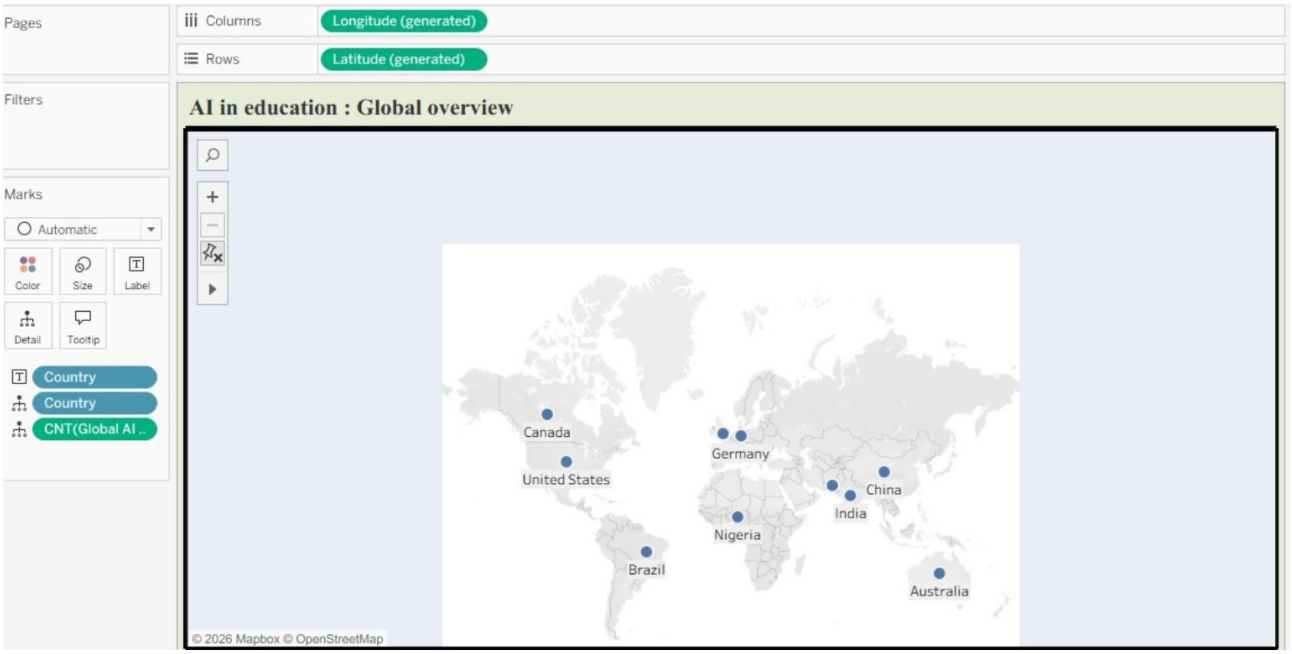


Figure 7. AI in education: global overview map.

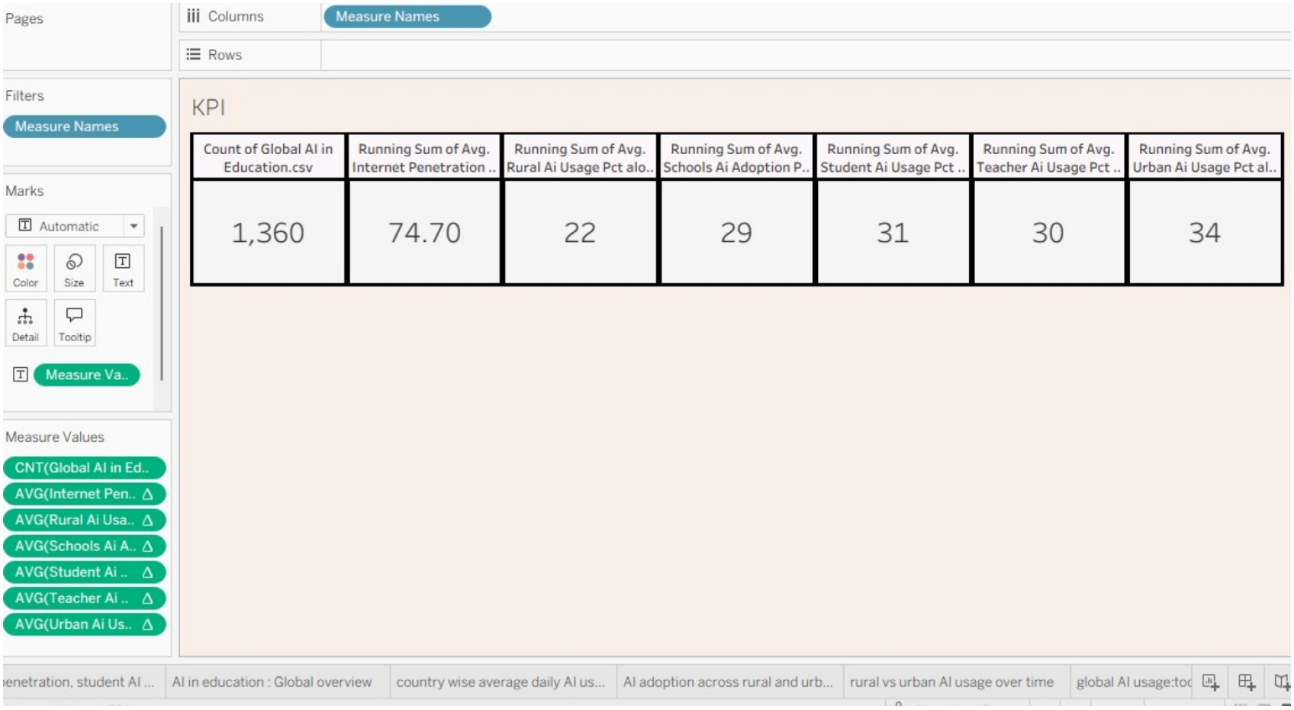


Figure 8. KPI view of the main project measures.

5. Dashboard

5.1. Dashboard Design File

Two Tableau dashboards were developed to combine the worksheets into a compact analytical view. The dashboards use KPI cards, country comparisons, a global map and adoption-level charts. The supplied screenshots are included below without altering their Tableau content.

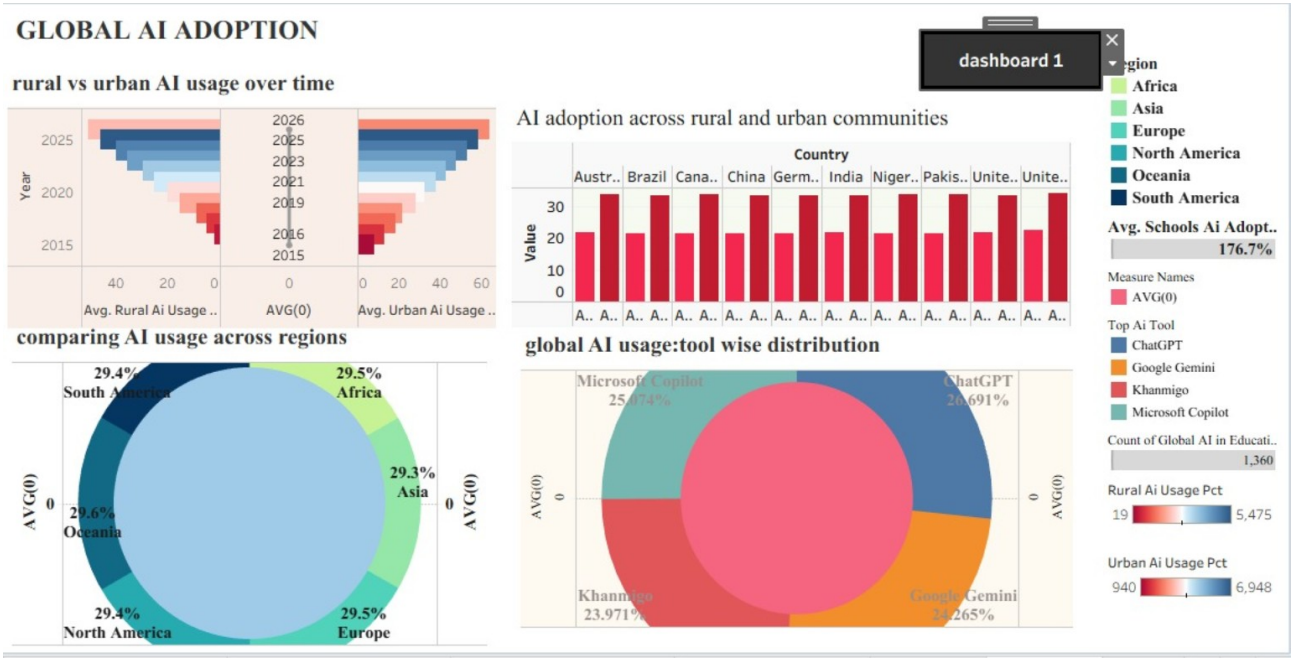


Figure 9. Dashboard 1 – Global AI Adoption.

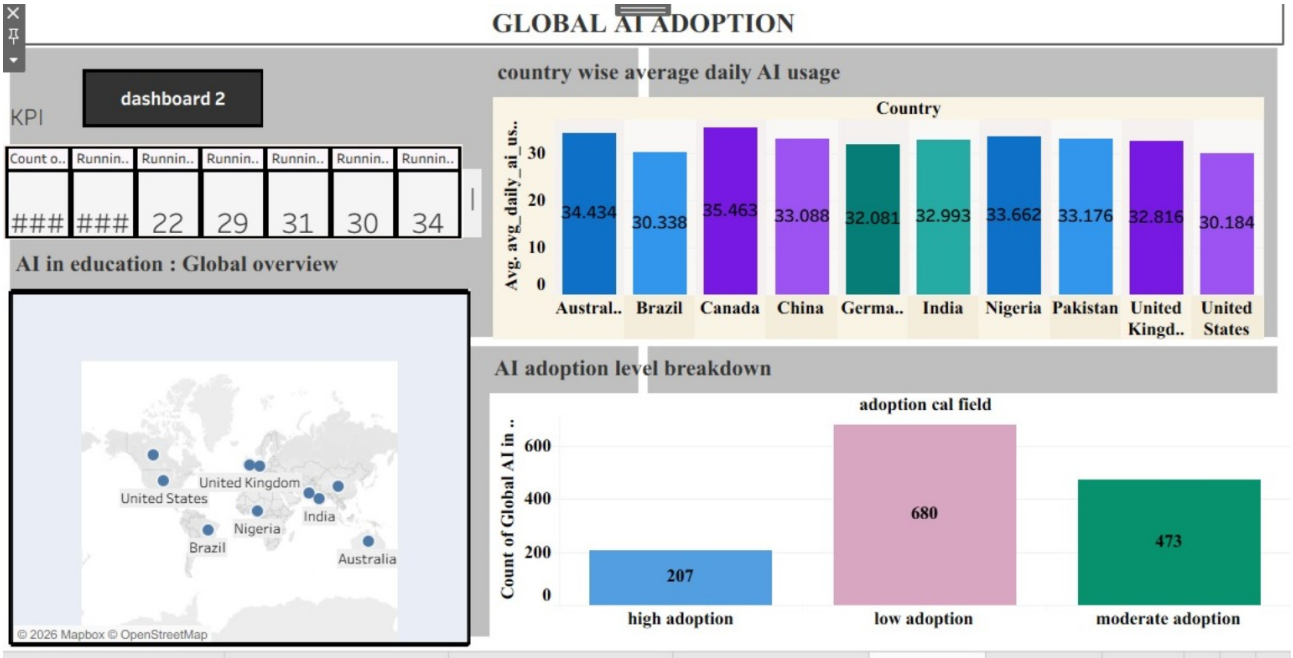


Figure 10. Dashboard 2 – Global AI Adoption.

Dashboard design principles applied include a clear layout, appropriate visualization types, consistent visual encoding, readable titles, KPI summaries and interactive Tableau filtering where available.

6. Report

6.1. Story Design File

The Tableau Story presents the analysis as a sequence of views so that the audience can move from country/community comparison to time trends, regional comparison, tool distribution and other supporting views.

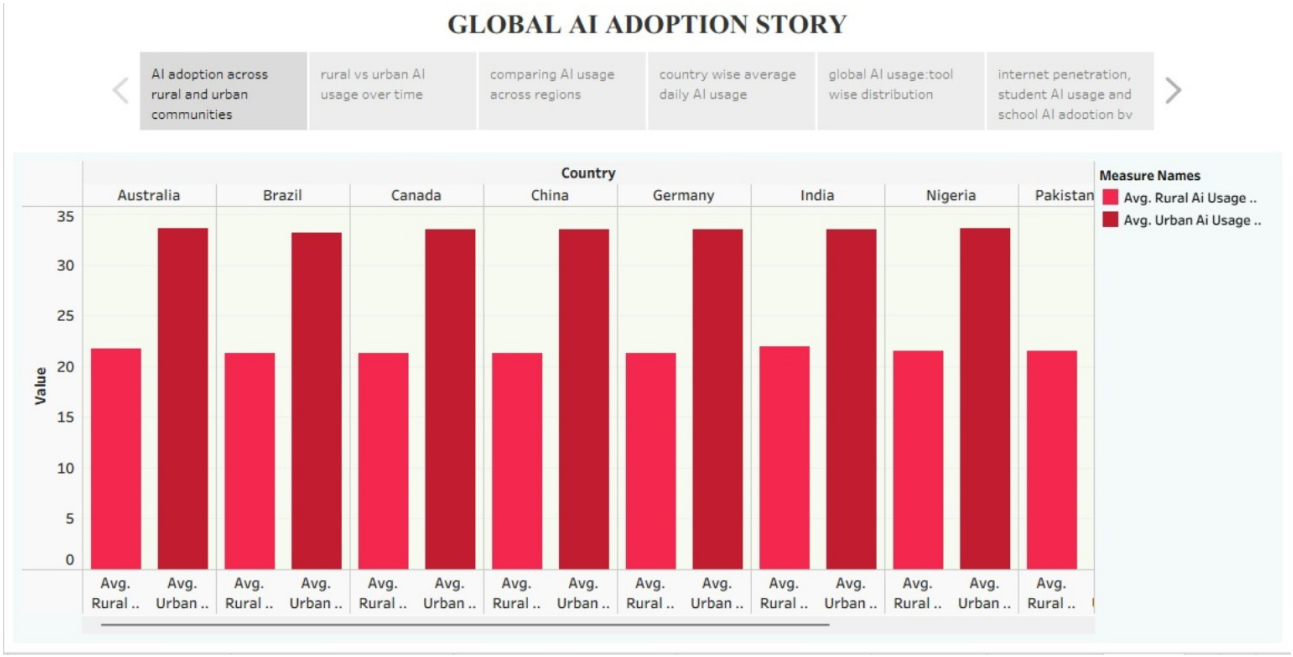


Figure 11. Tableau Story – Global AI Adoption Story.

The story observations include the rural/urban difference over time, adoption-level counts, AI tool distribution, country-level daily usage, regional comparison and the relationship between internet penetration and school AI adoption.

7. Performance Testing

7.1. Utilization of Data filters

The Tableau workbook uses dashboard/story controls and field-based filtering to allow the viewer to focus on relevant countries, regions, measures and views. The screenshots show country/measure legends and dashboard controls used for interactive exploration.

7.2. No of Calculation Field

The project uses calculated measures/categories for average daily AI usage, average rural/urban usage and adoption-level classification. The KPI worksheet also summarizes the principal measures used in the dashboard.

7.3. No of Visualization

Eight primary business-question visualizations were documented: rural vs urban usage over time, regional comparison, country-wise daily usage, adoption-level breakdown, AI tool distribution, internet penetration vs school adoption, rural/urban country comparison and the global daily-usage comparison. Additional KPI and map views support the dashboards.

Performance testing was performed as a functional Tableau review: filters/controls were checked for usability, calculated measures were used in the workbook, and the visualizations were combined into dashboards and a story.

8. Conclusion/Observation

- The dataset contains 1,360 records; the project adoption classification shows 207 high-adoption, 473 moderate-adoption and 680 low-adoption records.
- Average school AI adoption is close across regions, ranging approximately from 29.25% to 29.63%.
- AI tool distribution is relatively balanced; ChatGPT represents 26.69% of the records in the supplied analysis.
- Canada has the highest average daily AI usage at about 35.46 minutes among the countries shown, while the United States is about 30.18 minutes.
- Average urban AI usage is higher than average rural AI usage across the period shown.
- The project visualizations show a strong increase in AI usage over 2015–2026.

9. Future Scope

- Add more countries and longer time periods as new data becomes available.
- Incorporate more detailed internet infrastructure and affordability indicators.
- Add policy-specific analysis when complete and reliable policy data is available.
- Compare AI adoption by student and teacher groups in greater detail.
- Publish the Tableau dashboards and story on Tableau Public for interactive access.
- Add predictive or forecasting analysis in a future version, while keeping descriptive findings clearly separated from predictions.

10. Appendix

10.1. Source Code (if any)

No source code was provided in the submitted materials. The project was developed using the provided CSV dataset and Tableau visualizations.

10.2. GitHub & Project Demo Link

No GitHub repository or public Tableau demo link was provided in the submitted materials. These links can be added after publication.

Dataset: Global AI in Education.csv — project-provided source file.

Documentation template: Final Project Report Template, with sections 1–10 as supplied by the teacher.
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